**Nuo Xu** Telephone: (+86)185-0212-1466 Email: nxu8@outlook.com
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## EDUCATION BACKGROUND

**Johns Hopkins University** ***09/2017-05/2019***

Degree: Master of Science in Engineering Major: Electrical and Computer Engineering GPA: 3.73/4.00

**Tianjin University** ***09/2013-06/2017***

Degree: Bachelor of Engineering Major: Automation GPA: 3.71/4.00 Rank: 15/168

***Honors:*** *2017 Honor Graduate(10%); 2015 School-level Merit Student (10%);* 2016 *Outstanding Class Leader；2015 Zhoukou Chamber of Commerce Scholarship (1/112)*

## TECHNICAL SKILL

- Software: proficient in ROS, OpenCV, Git; familiar with Linux, TensorFlow, Pytorch
- Programming Skills: competent in Python, MATLAB; familiar with Java, C++
- Hardware: experienced in Arduino, PLC, Universal Robot, TurtleBot

## WORK EXPERIENCE

***Software Development Engineer*, Data Grand Tech Inc. (Shanghai, China)** ***06/2019-now***

- Developed hand-writing Chinese characters recognition system with CRNN which achieved above 90% correction
- Developed Bank-Card Recognition System which can recognize Chinese Bank Cards and Bank Account within 1000ms and over 90% correction.
  - Used PSENet and CTPN to detect the text location, and CRNN to recognize the text content
- Developed Table Detection System in Documentations
  - Applied FPN and Faster-RCNN on post-processing result which converted documentation to natural image for easier classification

## ACADEMIC & RESEARCH EXPERIENCE

**A Tape-cutting Procedure on the Orbiting Satellite Based on the DaVinci Surgical Robot** ***06-08/2018***

*Supervisor: Prof. Simon Leonard*

- Realized UR10 Cartesian path planner by picking up passing waypoints with DaVinci console in order to cut and peel back the plastic tape of the satellite
- Developed the Cartesian path planner with collision detection in *MoveIt!* in C++
- Achieved the goal of viewing the path generated from the planner in the stereo image with Rvinci plugin, RVIZ Interaction and Navigation with DaVinci, and picking points in RVIZ with the gripper of DaVinci from the stereo image

***Software Engineer*, Hopkins Delivery Project (*Sponsored by Whiting School of Engineering*)** ***09/2017-05/2018***

*GitHub: https://github.com/NormXU/Hopkins-Delivery; Demo Video: https://vimeo.com/256362144*

*Supervisor: Prof. Charbel Rizk*

- Demonstrated the concept of drone delivery system built upon DJI Matric 100, ZED stereo camera, NVIDIA Jetson TX2 and Arduino throughout the JHU campus
- Developed a ROS package for communication between ultrasound sonars, electromagnetics and Arduino & NVIDIA Jetson TX2 with rosserial
- Developed the landing algorithm that allows the drone to land on specified tag on the ground by detecting the AprilTags position with downward camera

**Visualization System of Acupuncture Manipulation based on Augmented Reality** ***02-06/2017***

- Developed the hardware on the FPGA with Verilog HDL
- Programmed the human acupoints registration algorithm in host computer with MATLAB
- Implemented the feature extraction and matching, using the Speeded-Up Robust Features (SURF), in the host computer which consists of the applications of 3D imaging and spatial affine transformation on the principle of computer geometry to realize the precise registration of human acupoints to the real human body that can be

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picked up by camera module

## *GRADUATE COURSE PROJECT*

**Deep Learning: Self-driving Guidance** ***03-05/2019***

*GitHub: https://github.com/NormXU/Self-Driving-Guidance*

- Achieved the goal of segmenting the road with U-Net+Vgg16, detecting the front car with YoLo V3, and estimating the distance in-between with DepthNet based only on the RGB camera

**Machine Learning: Collaboration System between Amazon Alexa and TurtleBot** ***08-12/2018***

*GitHub: https://github.com/NormXU/AlexaBot; Demo video: https://www.youtube.com/watch?v=TXflYI3eTqI&t=7s*

- Realized the functions of using hand gesture to control the TurtleBot remotely with the Convolutional Neural Network and Bayes classifier utilizing the deep learning framework of Keras, tracking the specific user through the Microsoft Kinect depth camera, and scene description with Dlib through training by SVM+HoG

**Computer Vision: Augmented Reality Drum Game Development with OpenCV and OpenGL** ***08-11/2018***

*GitHub: https://github.com/NormXU/AR_Drum; Demo Video: https://www.youtube.com/watch?v=bi6BYpFSOGs&t=24s*

- Identified the action of the drumstick through the color filters and frame differences, and determined the actual position of the virtual drum with AR Tag.

**Robotic Device: Pick-and-Mark and Draw Portrait with UR5** ***05/2018***

*Demo video: https://www.youtube.com/watch?v=OsZfEkdaMbs&t=4s*

- Programmed the robot to identify the position of the start and the target locations, pick up items and complete other types of control, like inverse kinematics, rate control using differential kinematics, and Gradient-based control, in order to draw the professor's face on paper consisting of more than 2,000 points

## *TEACHING ASSISTANT*

**Course: Algorithms for Sensor-based Robotics** ***08-12/2018***

- Assisted the professor in assigning and correcting homework for 20 graduate and undergraduate students
- Helped students to comprehend the concepts, like the hand eye calibration, obstacle avoidance and path finding based on the UR5 industrial mechanical arm in MATLAB, and the autonomous obstacle avoidance and SLAM based on TurtleBot in C++

## *PATENT*

**Patent:** Jiang Wang, Shuangming Yang**, Nuo Xu,** Huiyan Li, Bin Deng, Xile Wei, Chen Liu; *Visualization System for Acupuncture Manipulation Based on Augmented Reality and Its Image Processing Method;* cn 201710231105.1

## *CONPETETION*

***Algorithm Engineer*, IEEE International Conference on Robotics and Automation 2018 DJI RoboMaster AI Challenge** ***03-05/2018***

- Undertook the responsibilities of camera calibration and recognition algorithm development, which allows the robot to detect the enemy by HoG + SVM
- Authored a report: https://normxu.github.io/resource/Johns_Hopkins_University__Hopkins_AI_ICRA_Technical_Report-ilovepdf-compressed.pdf

## *PRACTICAL EXPERIENCE*

***Intern* (20 hours/week) Tianjin Career Training Center** ***08-09/2016***

- Designed a water tank control system with SIMENS PLC to maintain the water level at the setting level
- Built a system which can collect, transfer and classify adjunctions for grinding wheels with PLC

***Intern* (40 hours/week), Department of Microelectronics and Nanoelectronics, Tsinghua University** ***07-08/2016***

- Acquired coding knowledge in OpenCV for digital process, debugged and self-learned STM microchip and machine vision